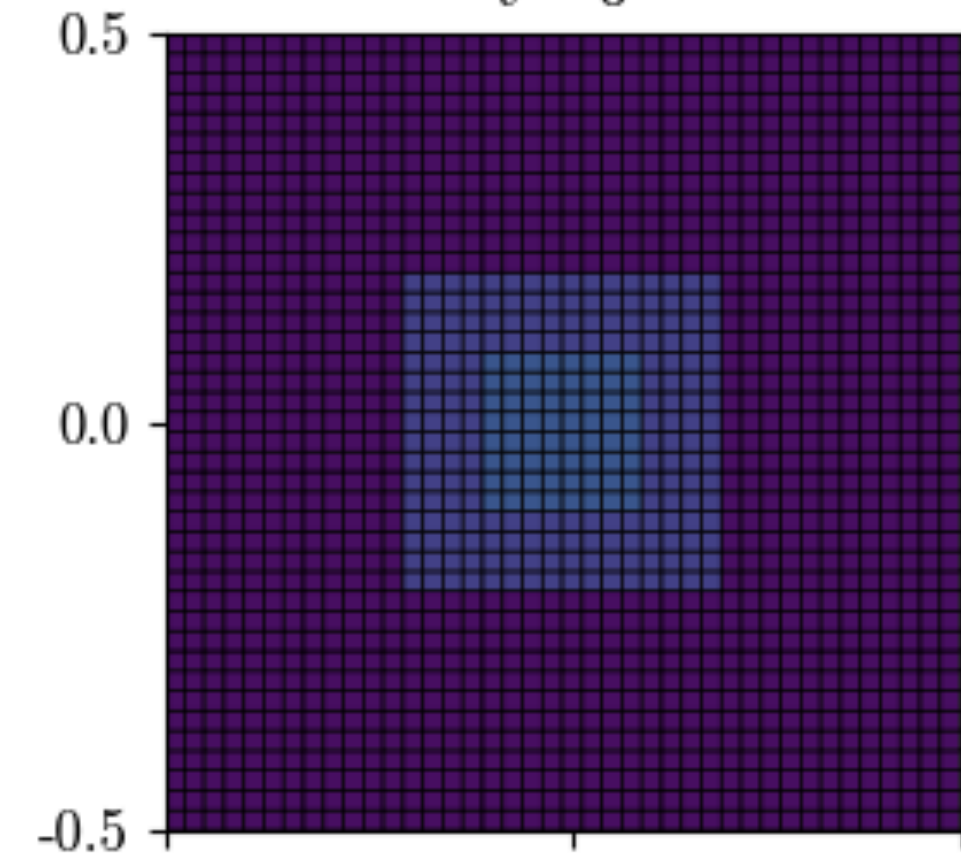
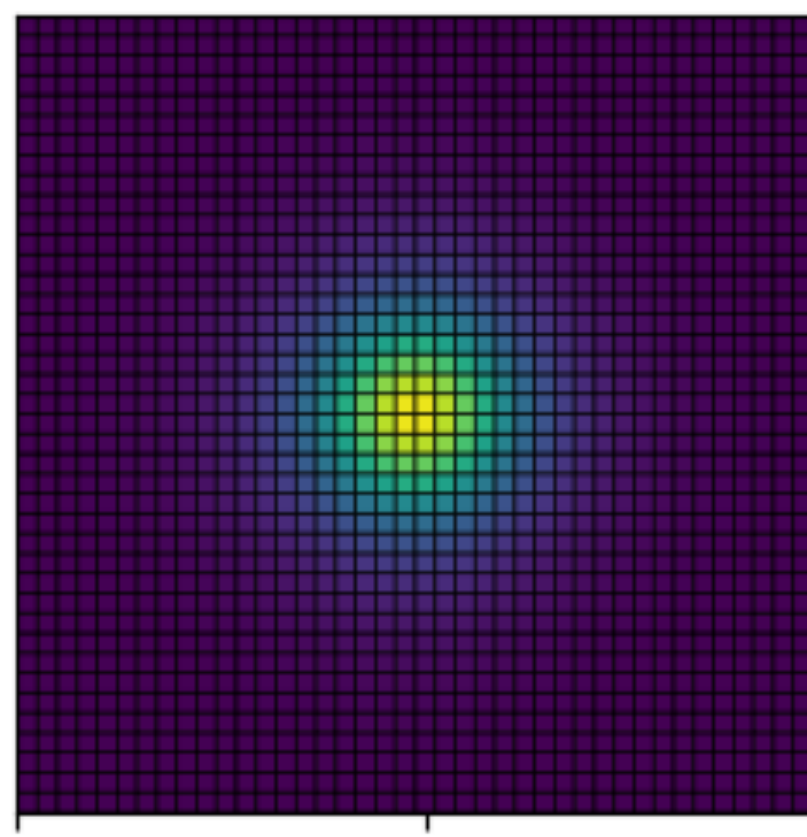


$$\bullet \quad \begin{cases} \partial_t f &= \sigma_x \Delta_x f - \text{Pe} \nabla_x \cdot (v(\theta) f) + \partial_\theta^2 f - \chi \partial_\theta (n(\theta) \cdot (\nabla K * \rho)_\tau f) \\ f(t=0) &= f_0 \end{cases}$$

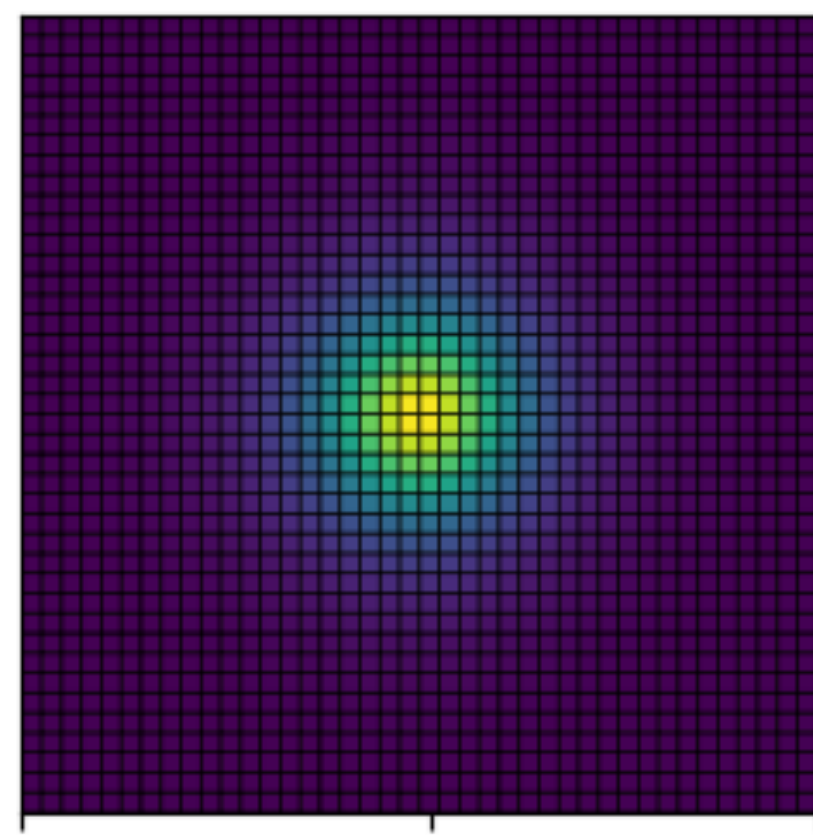
$t = 0$



$t = 1.5$



$t = 3$



ρ

12.5

10.0

7.5

5.0

2.5

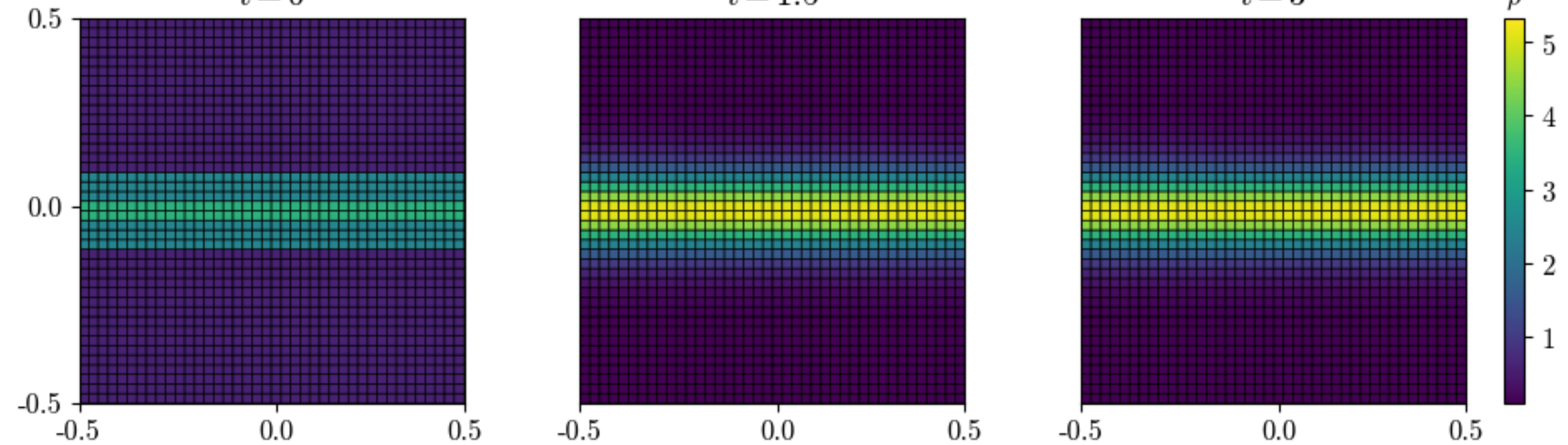
$$\tau = 0$$

$t = 0$

$t = 1.5$

$t = 3$

ρ



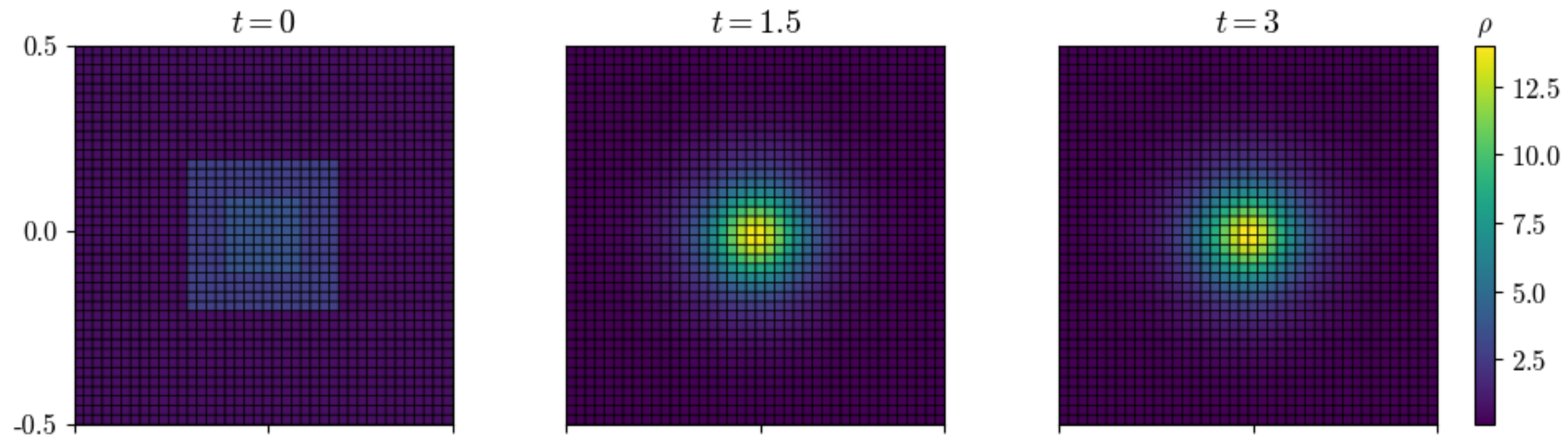
$$\tau = 0.2$$

Antdynamics

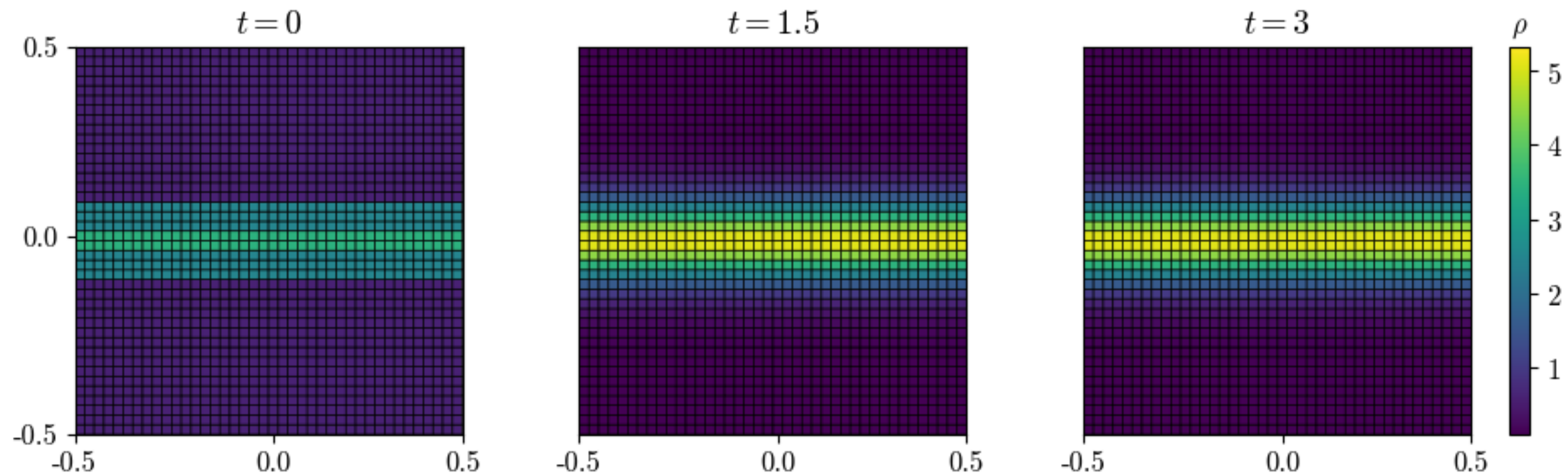
Ant dynamics

- $$\begin{cases} \partial_t f &= \sigma_x \Delta_x f - \text{Pe} \nabla_x \cdot (v(\theta) f) + \partial_\theta^2 f - \chi \partial_\theta (n(\theta) \cdot (\nabla K * \rho)_\tau f) \\ f(t=0) &= f_0 \end{cases}$$

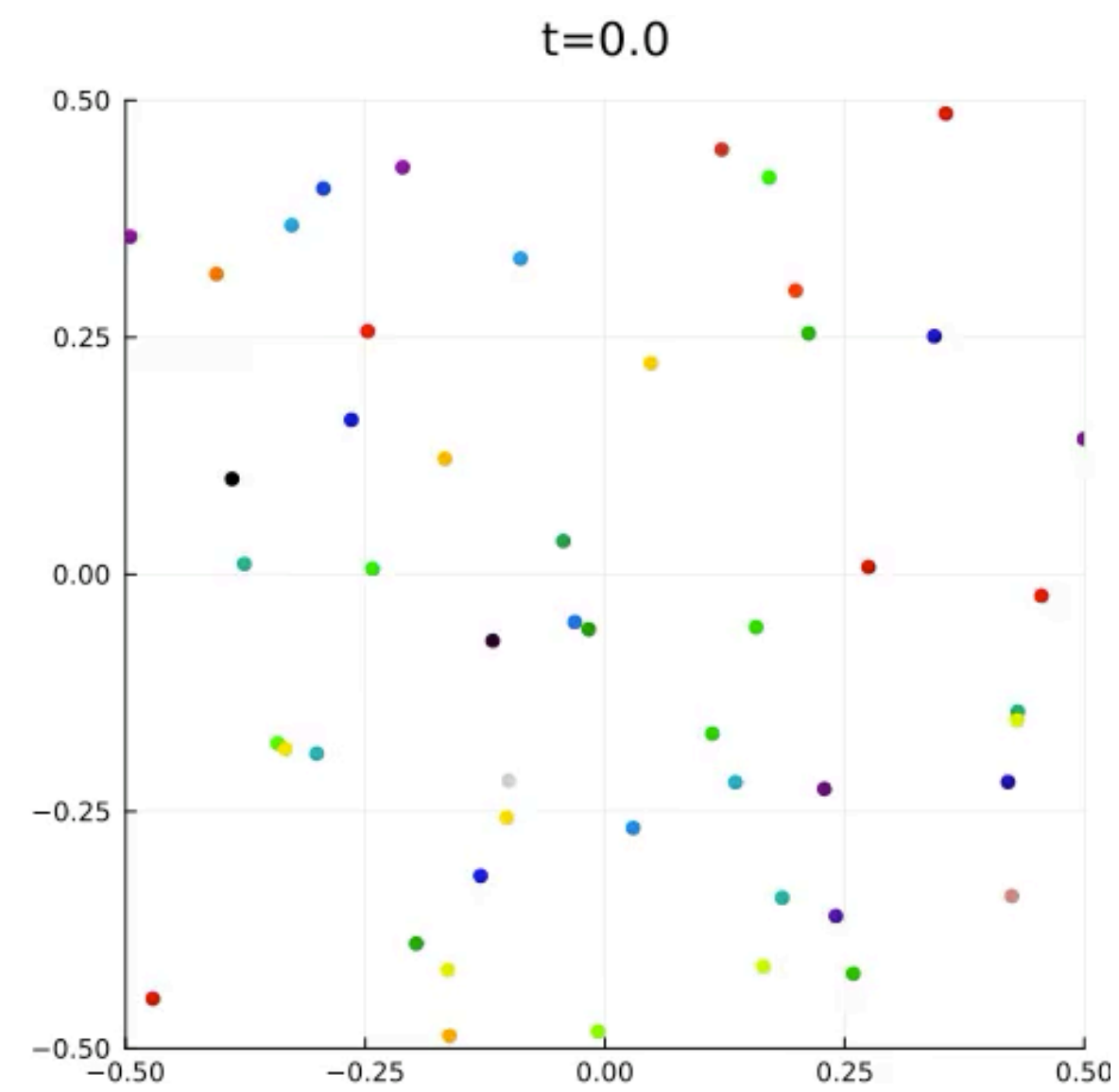
$\tau = 0$



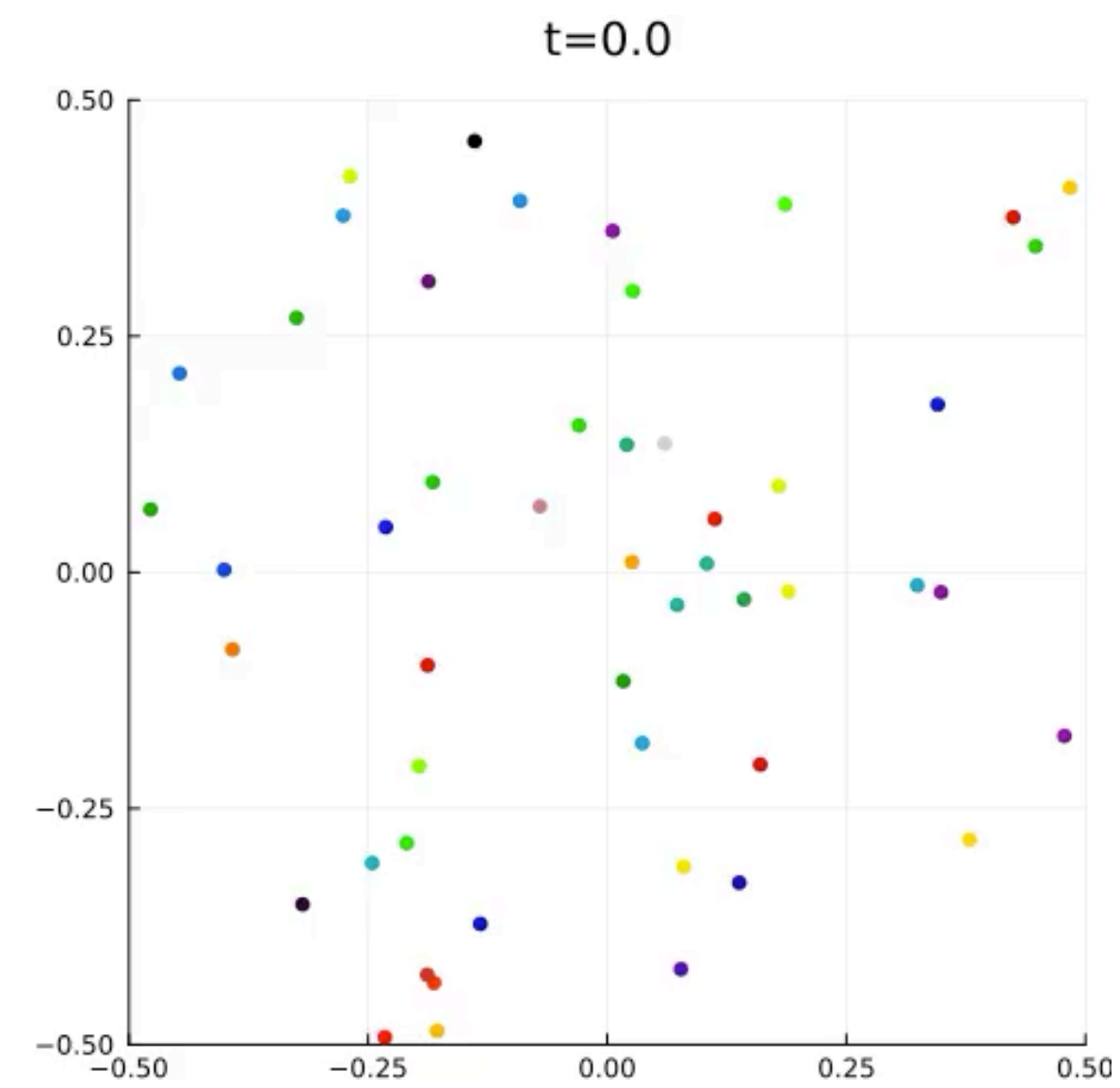
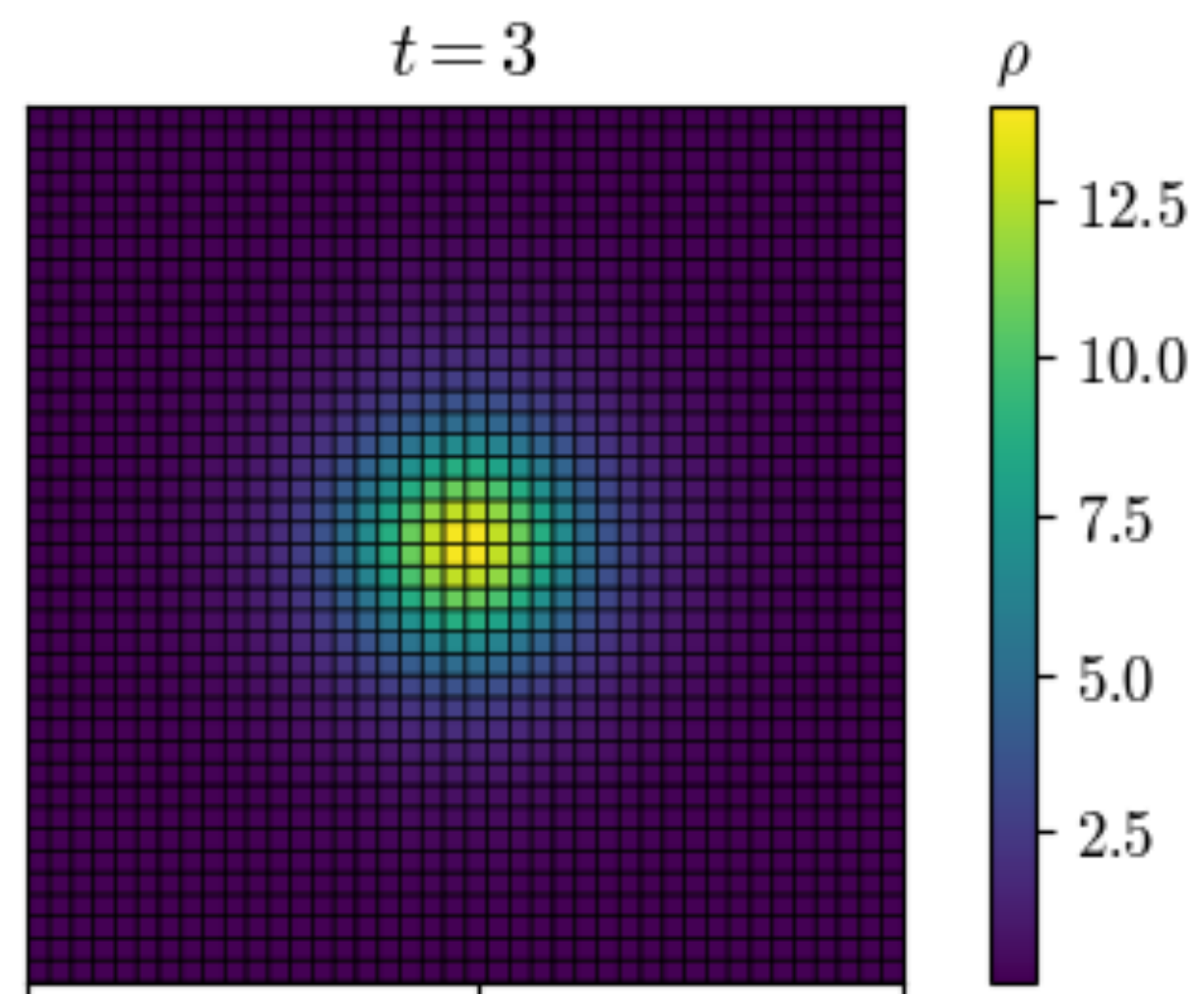
$\tau = 0.2$



Ant dynamics



$$\tau = 0$$



$$\tau = 0.15$$

